Advance Manufacturing Techniques

(Machining Processes & Operations)

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Enterprise Manufacturing Methodologies

- Lean Manufacturing: (5 Principles) Lean Enterprise Research Centre (LERC)
- Agile Manufacturing: (the 5C's, 4 key characteristics & the 7 Prerequisites)
- 'Just-in-Time' production:
- "Group Technology" concept.
- Flexible Manufacturing:
- CAD/CAM manufacturing:
- Computer Integrated Manufacturing (CIM):
- Computer-Aided Part Programming (CAPP):
- Six Sigma (Quality Contril & Quality Assurance)
- Co-ordinate Measuring Machines (CMM)
- Rapid Prototyping
- 'Live Tooling' & 'Automatic Tool Change'

Types of Machining Operations

- Traditional machining: (3000 BC Ancient Egypt)
- Non-traditional machining: (2300 BC Ancient Egypt)
- CNC machining: (1949 MIT, commercialised in the 1960's)
- AI-driven CNC machining (2014)

Process vs Operation

- A process involves more than on operations, while an operation is consist of a single activity.

Machines vs Devices

- Machines typically have engines or other drive mechanisms that enable movement and operations. Devices, on the other hand, may be manually operated or powered by simpler energy sources. Machines are often multifunctional, meaning they can integrate various processes and perform multiple operations simultaneously.

Machines vs Tools

- While tools focus on manual, precise tasks, equipment combines tools to tackle complex operations, and machines automate processes for maximum efficiency. Recognizing their differences helps in choosing the right resources for your needs.

Die Tool Manufacturing Process

- dies are a subset of tools—all dies are tools, but not all tools are dies.
- dies are specialized precision manufacturing machine-tools made for industry.
- Dies are Machine-tools that used in the mass production lines, assembly lines, processing & packaging lines or as a stand alone tool making machine/ tool.
- There are male and female dies, the male component is the protruding component, while the female is the depression or indentation.
- Die machine-tools are essential for mass production across various industries.
- the precision and consistency of modern products, from car parts to packaging materials, owe a great deal to die manufacturing process.

Die Types

- Simple Dies
- Compound Dies
- Progressive Dies
- Transfer Dies
- Multiple Dies

Die Parts/ Components

- Die block
- Die holder
- Punch plate
- Stripper plate
- Guide pin
- Pressure plates/ Back-up plates.

International Standardisation Bodies

- (BIPM) - Bureau International des Poids et Mesures.
- OIML - International Organization of Legal Metrology.
- ILAC - International Laboratory Accreditation Cooperation.
- IEC - International Electrotechnical Commission
- ISO - International Organisations for Standardisation
- SAE - Society of Automotive Engineers
- CEN(EN) - European Committee for Standardisation
- DIN - Deutsches Institut für Normung e.V. , the German Institute for Standardisation.
- BSI(BS) - British Standards Institute (mostly Metric with old Imperial),
- BS-EN-ISO/IEC - this prefix means, that the related standard is overseen and inspected by all four stndardisation institutions.
- ANSI - American National Standards Institute (Imperial).
- JIS - Japanese Industry Standards (Fully Metric since 1950's with some old Imperial standards still in use).

ISO – International Organization for Standardisation

- a global network of the world's leading standardisers.
- ISO has developed over (26,092) International Standards and other deliverables. All are included in the ISO Standards catalogue.
- a formula that describes the best way of doing something. It could be about making a product, managing a process, delivering a service or supplying materials – standards cover a huge range of activities.
- It starts with a collection of best practices, which evolves to a framework once adopted by large organisations then eventually become a standard when adopted and enforced by Governments and regulators.

ISO Standards for Manufacturing Industry

- TC - Technical Committee → (ISO/TC 45/SC 1)
- SC – Science Committee → (ISO/TC 45/SC 1)
- ICS - International Classification for Standards → (UID) e.g. (ICS 23.040.70).
- (ISO/IEC 80000-:2022) Quantities and units, Part 1: General.
- (ISO 9974-1:1996(en)) Connections for general use and fluid power — Ports and stud ends with ISO 261 threads with elastomeric or metal-to-metal sealing — Part 1: Threaded ports.
- (ISO 261:1998(en)) ISO general purpose metric screw threads — General plan
- (ISO 8434-1:2018) Metallic tube connections for fluid power and general use, part 1: 24° cone connectors. 24-degree taper fittings.

ISO Standards for Manufacturing Industry

- (ISO 18752:2025-06 Edition 5) Rubber hoses and hose assemblies.
- (ISO 6743-4:2015) Lubricants, industrial oils and related products (class L, Part 4, Family H, Hydraulic systems).
- (ISO 6149-1:2022) Connections for hydraulic fluid power and general use — Ports and stud ends with ISO 261 metric threads and O-ring sealing part 1: Ports with truncated housing for O-ring seal.
- (BS EN ISO 12151-2:2022) Connections for hydraulic fluid power & general use - Hose fittings, part 2: Hose fittings with (ISO 8434-1) 24° cone connector ends with O-rings.

SAE- Society of Automotive Engineers

- SAE is based on the metric system and the SI units.
- SAE is recognised around the world as a top tier standards development organisation for Automotive, Commercial-vehicle, and Aerospace standards.
- (SAE J517_202007) - Hydraulic Hose.

DIN - Deutsches Institut für Normung e.V.

- Represents the German Institute for Standardisation.
- DIN is based on the Metric system and the SI units.
- (DIN 2353:2013-01) Non-soldering compression fittings with cutting ring - Complete fittings and survey.

Unit Systems

- ***Metric System:*** (which is the SI International units system)
- ***Imperial System***: this is the post-1824 British-Imperial system, adopted by the USA, Liberia and Myanmar while partially used in the UK & Commonwealth countries.

Metric Measurement System

- It is the International standard system for measurement based on the SI-units.
- All ISO, DIN and SAE standards are based on the Metric system & SI units with conversion details to imperial whenever required.
- Units: (mm, cm, m, km, g, kg, Tone, Liter, Bar, sec., Pa, Rad, Hz, Kelvin, °C, Newton, Joule, Watt, Volt, Ampere, Tesla, Weber, Henry, Candela, mole, Metric-Dunam, Metric-Hectare & Int-inch)

Imperial Measurement System

- Units: [inch(in/”), Yard, Foot, Furlong, Mile, Grain, Stone, Ounce(Oz), Gallon, Pound, psi, Acre, BTU & Imperial-Dunam, Imperial-Hectare].
- (1/16, 1/8, 1/4, 3/8, 1/2, 5/8, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, & 4 in) fittings & sockets sizes.
- The “International-inch” was standardised by an international agreement on 1st July 1959 in terms of the metric system to 25.4mm.

Imperial Industry Standards

- **BSP (UK):**
- (BSP): British Standard Pipe
- (BSPT): British Standard Pipe Thread, this is a Tapered thread that features a 55-degree flank angle and narrows as it moves toward the end, forming a mechanical seal as it tightens.
- (BSPP): British Standard Parallel Pipe, a parallel (straight) thread from the BSP family, maintains the same diameter throughout.
- **NPT (USA):**
- NPT: (National Pipe Thread) (ANSI B1.20.1) a tapered thread, with a 60-degree thread angle.
- NPS: (National Pipe Straight-thread) (ANSI B1.20.3) a straight thread
- NPTF: (National Pipe Thread Fuel) The differences come in the major and minor diameters, the root and crest of the threads. thread roots are designed to interfere with the crest of the mating thread with the intention of creating a mechanical seal through thread form deformation at assembly.
- JIS: (Japanese Industrial Standard) The Old JIS standards (pre-dating 1950) were based on some old imperial-Japan standards, while some standards were influenced by the British imperial system, where thread & fittings used 30-degree flare angle and BSP threads.

Workpiece (Stock Material)

- is the initial piece of raw material, such as a bar, rod, plate, or sheet, from which a final part is produced by cutting away excess material. Choosing the correct stock is crucial and depends on the desired final shape, the type of machine, the material properties needed, and the cutting tools available. Stock is often a pre-formed shape that includes extra material, called "machining allowance," which is removed to achieve the final dimensions and a high-quality surface finish.

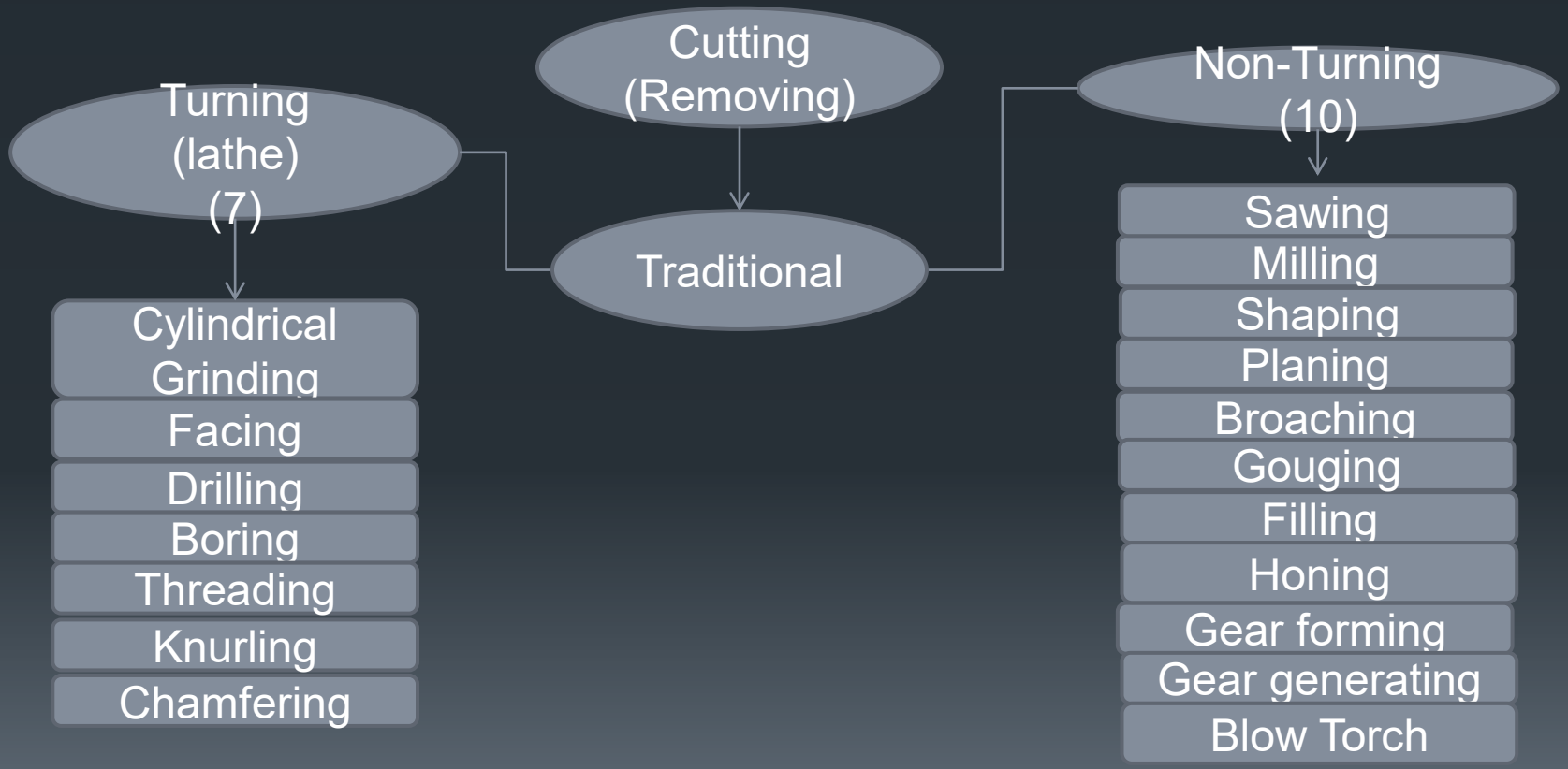
Taper Thread

- A taper thread is a type of thread that has a slight taper along its length, meaning the diameter of the thread gradually increases or decreases from one end to the other. This design creates a tight seal when the male and female threads are joined, making it ideal for applications where leak-tightness is crucial.

Machining Processes

- ***Removing(Cutting)***: This process involves the use of cutting techniques & tools to remove excess materials from a workpiece on direct contact. This machining operation occurs when the cutting tool directly interacts with the workpiece.
- ***Treating:*** [Annealing, Hardening & Coating].
- ***Forging***: (Hot-Shaping).
- ***Forming***: (Cold-Shaping) [Pressing, Knurling & Rolling].
- **Casting & Molding**:
- ***Adding***: [Welding, Stress fit, Interference fit, Joining & Galling].

Cutting Process (Operations)



Cutting Process (Removing Process) (Tools & Techniques)

- Traditional: (Circular & Non-Circular)
- Non-traditional: (AWJM, LBM, PBM, USM, etc.)

Traditional Cutting Operations

Turning (Lathe) Operations

- In the turning process, the cutting tool remains stationary while the workpiece rotates. It is a lathe operation and involves the removal of excess material from a workpiece with the aid of a cutting tool. Besides, the cutting tool works along two axes of motion, creating cuts with precise width and depth.
- Additionally, the turning process works great for machining the interior or exterior part of a material. Turning performed on the material exterior part is known as facing, while that done on the inside is known as boring.

Facing Operation

- is a machining process that creates a flat surface on the end of a workpiece, either by rotating the workpiece against a stationary cutting tool on a lathe or by rotating a cutting tool against a stationary workpiece on a milling machine. This process is used to shorten a workpiece to a precise length, make the end perpendicular to its axis, and provide a smooth, flat reference surface for subsequent machining.

Drilling Operation

- Drilling involves using drill bits (multi-point cutting tools) to produce cylindrical holes in solid materials.
- Drill bits used for this procedure feature two spiral channels. These channels, also known as flutes, evacuate the “swarf” or chips out of the hole as the drill bit progresses into the material.
- The holes created by this drill press often aid part assembly. Besides, drilling is applied first before tapping, reaming, or boring to create threaded holes or bring the dimension of a hole within an acceptable tolerance.
- This makes this operation one of the most important out of the other types of the machining process.

Drilling Tools

- Lathe drilling
- CNC drilling
- Mill drilling
- Bench drill (Pillar drill): Vertical drilling
- Magnetic drill: Vertical or Horizontal drilling (mobile drill)
- Handheld drill: Battery, wired or Air powered (mobile drill)

Boring Operation

- is a machining process used to enlarge and refine an existing hole, making it more accurate than drilling alone. It uses a single-point cutting tool, or a boring head with multiple tools, to create or improve the internal diameter of a workpiece, achieving high accuracy and a good surface finish. This process is essentially the internal equivalent of turning and can be performed on machines like lathes and mills.

Threading Operation

- is a machining process to create helical grooves (threads) on a cylindrical workpiece either internally or externally and internally on a non-cylindrical workpiece. This is done by using a single-point cutting tool that moves linearly while the workpiece rotates, requiring precise synchronization between the tool's path and the workpiece's rotation. This process is essential for creating parts like bolts, screws, and nuts that require secure fastening.

Chamfering Operation

- is a machining process that creates a flat, angled surface on a sharp edge, most often at a 45-degree angle. This operation removes sharp corners, burrs, and stress points to improve safety, facilitate assembly, and enhance the aesthetic appeal of a workpiece. The process can be performed manually with tools like a file, or on a machine using specialized chamfering tools, making it a common technique in industries from automotive to aerospace.

Lathe Machining (Turning Ops.)

- In the turning process, the cutting tool remains stationary while the workpiece rotates. It is a lathe operation and involves the removal of excess material from a workpiece with the aid of a cutting tool. Besides, the cutting tool works along two axes of motion, creating cuts with precise width and depth.
- the turning process works great for machining the interior or exterior part of a material. Turning performed on the material exterior part is known as “facing”, while that done on the inside is known as “boring”.
- Can produce (flat, cylindrical & conical) surfaces.

Types of Lathe Turning Machine

(Bench or floor mounted)

- *Centre lathe*: uses a tailstock, was driven by a central steam engine
- *Engine lathe*: uses a tailstock, was driven by a steam engine
- *Turret lathe*: has stationary (fixed) hexagonal turret-head, while accommodate heavier work.
- *Capstan lathe*: has “sliding” hexagonal turret-head (tool block).

Both Turret & Capstan lathes are used for mass production and repeated jobs, while Center lathe & Engine lathe are used for one off job or simple jobs with less tool changing requirements.

Lathe Parts

- Headstock: contains the gearbox and where the shaft & Chuck are connected.
- Feed Gearbox & feed shaft.
- Tailstock: a single tool holder.
- Quill: The moving out part of the Tailstock.
- Turret-Head: a six tools indexed fixation (Tailstock replacement).
- Live-Centre: bearing loaded tool mounted on the Tailstock (rotates with the workpiece).
- Dead-Centre: bearing-less tool mounted on the Tailstock (no rotation).
- Chuck: (3-Jaws & 4-Jaws).
- Carriage: moves the tool-post on the z-axis (left & right), parallel to the shaft.
- Slide: (Cross-slide, Compound-slide & Turret-Auxiliary-slide)
- Tool-Post: the Cross-Slide carries 5 tools:
  (a front 4-way Tool-Post + a rear one-way Tool-Post) moves on the (x, z) axis.
- Moving Wedge: (The Tool-post handle)
- Compound-rest:
- Hand wheels: the x, z axis, Compound-rest & tail-stock wheels.
- Lead-Screw & Feed-Rod:
- Bed & Leadscrew
- Saddle: (bed-saddle & Turret-Saddle).
- Power feed: (Manual, Semi-Auto) & emergency stop switch
- Digital Read-Out (DRO).
- Grear levers
- Motor lever
- Coolant supply.

Power Feed System

- It is an automated mechanism, driven by an electric motor, that controls the motion of machine tools such as lathes and milling machines to achieve consistent and accurate feed rates.

Slides System

- *Cross-Slide:* moves the cutting tool across the (x-axis) perpendicular to the lathe's bed axis (z-axis).
- *Compound-Slide (Compound-Rest):* is a smaller, secondary slide that sits atop the cross slide of a lathe and rotates the tool-post at any angle to the lathe bed axis (not only perpendicular).
- *Turret-Slide*: in (Capstan lathe), it slides the Turret-Head longitudinally on the turret-saddle and along the bed axis (z-axis).

Engine Lathe

- a lathe machine with “Compound Slide” and Semi-automatic “power feed”.
- is a general-purpose horizontal lathe, either bench- or floor-mounted, equipped with a carriage and cross-slide, and often with semi-automatic power feed.

Center Lathe

- Equipped with: Cross-Slide or Compound-Slid & Tailstock (NO Turret) .

Non-Turning Operation

List of Workshop Tools

- Sawing Operation (sawing machines)
- Milling Operation (Milling Machines)
- Honing Operation (Honing Machines)
- Grinding Operation (Grinding machines)
- Drilling Operation (Drilling machines)
- Pressing Operation (Pressing machines)
- Blow-Torch Operation (Blow-Torch Device)
- Gouging Operation (Gouging Device)

Press Cutting Operation
(Press Machine)

- A press machine is a manufacturing tool that uses high pressure, typically from a ram or die, to change the shape of a workpiece through methods like (forming, cutting, or stamping). These machines are powered by various mechanisms, including hydraulic, mechanical, pneumatic, or servo systems, and are used in many industries to work with materials like metal, wood, and composites.

Sawing

(Wheel saw, Band saw, Chain saw).

- Sawing aims to create shorter lengths from extruded shapes, bars, and other materials, performed by cut-off machines. There are different cut-off machines engineers use for sawing, including; power hack saws, circular saws, and abrasive wheel saws.
- Besides, the saw band speed for sawing depends on the material. For instance, softer materials like aluminum alloys require a cutting speed of 30m/min or more. On the other hand, some high-temperature alloys require a slower cutting speed of 10m/min.

Milling Operation
(Vertical & Horizontal)

- Milling is a machining process that involves the use of rotating cutters to remove material from a workpiece. There are two main types of milling operations; face milling, slab milling and Fly cutting.
- Face milling is a machining manufacturing process used for smoothing or flattening the surface of workpieces. slab milling is ideal for machining a wide flat surface. It also is used for making heavier deeper cuts across a workpiece for fast material removal.
- Fly cutting uses 1 insert go produce a smooth surface finish.

Milling Machine Parts

- Motor, with switch
- Tool Head, with & variable speed control
- Quill
- Quill feed rate selection
- Quill feed handwheel
- Quill feed hand lever
- Spindle
- Digital readout
- RAM & RAM locks
- Ram adjust handle
- Table, with table stop dogs.
- Table locks
- Table traverse crank handle
- Table power feed
- Saddle, & saddle traverse crank
- Column
- Knee, & Knee vertical traverse crank
- Base

Honing Operation
(Vertical & Horizontal)

- is a precision machining process that uses abrasive stones to refine the surface finish and geometry of a workpiece, primarily used to achieve tight tolerances and a superior surface quality. It involves rotating and reciprocating an abrasive tool (or "stone") with controlled pressure inside a bore, which creates a characteristic cross-hatch pattern that is crucial for applications like engine cylinders to hold lubrication.

Grinding

- Grinding is one of the types of machining process ideal for improving the finish on a machined part's surface and tightening its tolerance. The process produces parts with identical shapes, finishes, and sizes. It is also the first step in further finishing operations like honing, lapping and superfinishing.
- Additionally, there are two main types of grinders:

1. *Surface grinders*: deals with flat surfaces.
2. *Cylindrical grinders:* (lathe grinding) both the cylindrical workpiece and the grinder abrasive wheel turns relative to each other, removing the excessive material for a finer surface finish.

Grinding Tool Types

1. Handheld grinder
2. Bench grinder
3. Belt grinder
4. Stone grinder
5. Angle grinder
6. Pedestal grinder
7. Precision grinder
8. Jig grinder
9. Gear grinder
10. Die grinder
11. Bore grinder
12. Surface grinder
13. Cylindrical grinder
14. Flexible grinder
15. Universal grinder
16. Center grinder
17. Center-less grinder
18. Tool & Cutter grinder
19. Form grinder
20. Plunge-cut grinder
21. Brush grinder

Blow Torch Operation

- Oxy–fuel cutting, a torch is used to heat metal to its kindling temperature. A stream of oxygen is then trained on the metal, burning it into a metal oxide that flows out of the kerf as dross.
- uses a mix of oxygen with “Acetylene or MAPP” gas, while smaller/home torches use either “butane or propane” gas.

Gouging Operation

- Thermal gouging is an essential part of welding fabrication, used for rapid removal of unwanted metal. The material is locally heated and molten metal ejected - usually by blowing it away. Normal oxy-fuel gas or arc processes can be used to produce rapid melting and metal removal.

Non-Traditional Cutting Operations

1. Electric discharge machines (EDM): (Plasma cutting/welding)
2. Plasma Beam Machining (PBM)
3. Laser Beam Machining (LBM): (Metal sheets, steel, stainless-steel, Aluminum, Copper, Brass, Titanium).
4. Water-Jet Machining (WJM): (a non-traditional machining process)
5. Abrasive Water-Jet Machining (AJM or AWJM): (Ultra high pressure).
6. Chemical Machining (CHM)
7. Electro-Chemical Machining (ECM)
8. Electro-Chemical Grinding (ECG)
9. Ultra-Sonic Machining (USM)

Water-Jet (WJ) Cutting

(used for cutting materials difficult to cut, such as)

- natural-stones: (Granite, Marble, Lime-stone, Sand-stone);
- Ceramic: as a transformed-material;
- manmade-stones [Porcelain (made of kaolin clay), Dekton, Neolith, Sile-stone, Ceasar-stone];
- PVC, plastic, rubber, foam, insulation/HVAC;
- glass, fiber-glass;
- non-Ferrous metals;
- Ferrous metals (Steel & Stainless-steel);
- composites;
- cardboard /paper;
- food.
- cutting electronics waist using (AWJ) .

Abrasive Waterjet (AWJ)

- In the AWJ machining process, the workpiece material is removed by the action of a high-velocity jet of water mixed with “abrasive” particles based on the principle of “erosion” of the material upon which the waterjet hits.

Electric discharge machines (EDM)
(plasma cutters)

- Use high-speed electrodes to melt or cut through metal materials. A plasma cutter can burn through various metals with varying degrees of speed and thickness. Metal fabrication shops, machine shops and even homeowner workshops will find this is the tool for them.
- EDM uses electrical discharges to cut through metal. EDM is an accurate method for cutting materials of 12 inches or less and often requires little-to-no secondary finish. Since it uses electricity, EDM is only compatible with conductive materials, including alloys and steel.
- The most common EDM fluid is (deionised water), which has had all of its mineral ions removed. This creates an environment with very low electrical conductivity.
- Used in Aerospace Industry.

EDM Limitations (CONS)

- Only works with electrically conductive material.
- Processing time is long.
- Electrodes need to be prepared.
- The direction of processing is limited.
- A heat-affected layer different from the grinding process will form.

EDM Used for Cutting & Welding Metals

- Military Industry
- Aerospace Industry
- Jewellery Manufacturing
- Pharmaceutical Manufacturing
- Automotive Production Process
- Surface Tension Testing
- Precious Metal Refining
- Power Tool Production

Treating Process (Operations)

(Involves Chemical, Electrochemical & Thermal processes)

- *Annealing:* Heating and slow cooling of a material that alters the properties to increase ductility (malleability) and reduce hardness, making it more workable or to increase toughness.
- *Hardening:* controlled heating to a critical temperature dictated by the type of steel (in the range 760-1300 C) followed by controlled cooling.
- *Coating:* such as Zinc coating, provide cathodic corrosion protection, i.e., when exposed to air and water, the zinc forms zinc carbonate, which protects the steel beneath against corrosion.

Forging process

(Hot Shaping)

It is the manufacturing process of hammering, forming metal into shape. This is either delivered from a hammer or a die. It is essentially the art of heating and working hot metal to design or shape that is fit for a specific purpose or use.

- Hammer Tool
- Die Tool

Forming Process

- Rolling Operation (Thinning)
- Knurling Operation (Stamping)
- Pressing Operation (Cold Shaping)

Rolling Operation
(Thinning)

- is a fundamental process in metalworking that plays a crucial role in bending, shaping and machining metal parts and workpieces. It is used to reduce the thickness of a material and give it a specific shape
- During smooth rolling on a lathe, you should tend towards a moderate to low rotational speed. The reason is simple: if the speed is too high, there may be vibrations on the workpiece that adversely affect the surface quality. Tip: as a rule of thumb, 200 to 400 rpm is a good starting value.

Knurling Operation

(Stamping/ Material Shifting)

- is a Turning machining process that produces a patterned texture on the surface of a (typically cylindrical) part by pressing or cutting a hardened tool into the material.
- Knurling is performed on a lathe machine.

Pressing Operation

(Cold Shaping / Pressure Shaping)

- A press machine is a manufacturing tool that uses high pressure, typically from a ram or die, to change the shape of a workpiece through methods like (forming, cutting, or stamping). These machines are powered by various mechanisms, including hydraulic, mechanical, pneumatic, or servo systems, and are used in many industries to work with materials like metal, wood, and composites.

Casting vs Molding(Moulding)

- Molding is for plastic and Casting is for metal. These processes are somewhat similar, though there are a few key differences to consider.

Casting Process (Operations)

- Sand Casting
- Die Casting
- Gravity Die Casting
- Pressure Die Casting
- Investment Casting
- Plaster Casting
- Lost Foam Casting
- Vacuum Casting
- Continues Casting

Molding Process (Moulding)

- Permanent Mold

Adding Process (Operations)

- *Welding:* (TIG, MIG, Flux, Stick, Spot, Gas, Brazing, Soldering)
- *Stress fit (Press fit): a form of fastening between two tightfitting mating parts that produces a joint which is held together by friction after the parts are pushed together.*
- *Interference fit:* (Heating & Shrinking by freezing or with liquid nitrogen)
- *Joining:* (Bolts & nuts, Screwing, Nailing, Riveting or interlocking).

Welding Operations

- *TIG welding:* uses a shielding gas, a non-consumable electrode and a hand-held filler rod to form the weld.
- *MIG welding:* uses a shielding gas and a machine-fed solid wire.
- *Stick welding:* relies on flux to release the shielding gas.
- *Spot Welding:* had an Ultimate Tensile Strength of 3261Kg compared to 1294Kg for an 8mm MIG plug weld
- *Flux-cored-Arc welding:* relies on flux to release the shielding gas.
- *Brazing: 450*°C melted added material fused to the workpiece material, ideal for fixing cracks in (Cast-Iron).
- *Oxyacetylene welding (Gas welding):*
- *Soldering:* similar to the Brazing process with lower temperature.

- The electric power supply used for welding is mostly DC or could be AC .
- *Shielding-gas:* uses different (%) mixture of Argon & Oxygen gases to provide the isolated atmosphere around the weld puddle.
- *Flux:* protects the weld pool from interacting with elements in the air during the welding process. Prevents oxides from building up while the material is heated.

TIG Welding

- is one of the most complex welding methods.
- uses a shielding gas, a non-consumable electrode and a hand-held filler rod to form the weld.

MIG Welding

- uses a shielding gas and a machine-fed solid wire.
- AC is hardly ever used in MIG welding.

Stick Welding
(informal)

- Shielded Metal Arc Welding (SMAW), a.k.a
- Manual Metal Arc Welding (MMAW or MMA), a.k.a
- Flux Shielded Arc Welding (FSAW).

is a gasless manual arc welding process that uses a consumable electrode covered with a flux to lay the weld.

Spot Welding

- is a form of resistance welding that joins together two or more metals (usually sheets of metal) by applying pressure and heat. The heat is generated by an electrical current. This is directed at specific, localised 'spots' rather than having a continuously moving weld – hence the name spot welding.
- the spot weld had an Ultimate Tensile Strength of 3261Kg compared to 1294Kg for an 8mm MIG plug weld

Brazing

- Brazing is a metal-joining process where a filler metal melts and flows into a joint between two or more base metals, joining them permanently. The filler metal has a melting point above (450°C) but below the melting point of the base metals, which are not melted during the process. This temperature difference is what distinguishes brazing from welding. Used specifically for Cast-Iron cracks welding.
- A flux is often used to clean the surfaces of oxides, preventing contamination and allowing the filler metal to wet the joint properly. The flux residue should be removed after brazing to prevent corrosion.

Brazing Applications

- Joining metals in engineering and manufacturing.
- HVAC applications (e.g., joining copper pipes).
- Repairing tools and components.
- Joining dissimilar metals that might be difficult to weld, such as Cast-iron.

Soldering

- Is more like Brazing, however the key difference is temperature. Soldering uses filler metal with a melting point below (450°C) and below the welded material melting point.

Galling (cold welding)

- Thread galling, an accidental welding, is a form of adhesive wear which occurs during the movement of components and often leads to stuck Nuts and seized Bolts. There are several causes of thread galling and ways you can prevent it.
- Materials which are highly ductile or which possess low work-hardening rates tend to be prone to galling. Austenitic stainless steels show a tendency to gall under certain conditions.

How to Prevent Galling

- Galling is a form of wear caused by adhesion between sliding surfaces. When a material galls, some of it is pulled with the contacting surface, especially if there is a large amount of force compressing the surfaces together. Galling is caused by a combination of friction and adhesion between the surfaces, followed by slipping and tearing of crystal structure beneath the surface. This will generally leave some material stuck or even friction welded to the adjacent surface, whereas the galled material will appear worn, chipped, or even gouged and may have balled-up or torn lumps of material stuck to its surface.
- To prevent it, use anti-cease lube on the threads before tightening them.
- Stainless-steal threads are notorious for galling. The wider the diameter of the thread and the faster the thread tightening process the higher the chance of galling.

Modern Machining

Computer Numerical Control (CNC)

- Modern Machining tools are equipped with CNC for automation, time saving, improved quality, reduced waist, higher efficiency, precision machining and reduced cost for mass production.
- All traditional and non-traditional machines mentioned previously can be CNC operated.
- With AI Assisted or operated CNC machines new levels of precision high quality mass production are achieved and new possibilities are unlocked.

Types of CNC Machines

- Mill CNC:
- Lathe CNC:
- Drilling CNC:
- Grinding CNC:
- Laser Cutting CNC:
- Plasma Cutting CNC:
- Waterjet Cutting CNC:
- EDM cutting/welding CNC:
- etc.

Type of CNC Operating Systems

- Real-Time Operating Systems (RTOS)
- Proprietary Embedded OS: Major manufacturers like FANUC, Siemens (Sinumerik), Heidenhain, and Mitsubishi use their own highly specialized embedded operating systems designed specifically for their machine control units (MCU).
- Mach3/Mach4: Widely used software that runs on Windows to convert G-code into motion commands. It is a staple for desktop and home-built CNC mills.
- GRBL: A high-performance, open-source firmware that runs on microcontrollers like Arduino. It interprets G-code and is frequently used in small CNC routers and laser cutters.
- LinuxCNC: A popular open-source system that runs on Linux (often Debian or Ubuntu) with real-time extensions. It can control a wide range of machines, from 3D printers to robot arms.
- IBM OS2-Warp-v4.52(32bit) final release replaced by its 3rd party replacement “ArcaOS”(32bit) sold by IBM, IBM Enterprise Systems Z/OS(Unix-based) (32bit)
- MS-DOS (16/32bit)
- MS-Windows (often Windows Embedded or IoT) 7/10/11 (32/64bit)

CNC Software Developers

- Motion Control
- G-Code & M-Code Generations
- Interface with CNC Hardware (CAD/CAM)
- System Security
- Software Updates
- Backup and Recovery
- Mastercam is well established and widely deployed among SolidCAM and Fusion 360 software.

CNC Software Categories

- CAD (Computer-Aided Design): Used to create the 2D or 3D digital design (e.g., AutoCAD, SolidWorks, Fusion 360).
- CAM (Computer-Aided Manufacturing): Translates the design into G-code (e.g., Mastercam, Vectric).
- Control Software: Reads the G-code and physically moves the machine (e.g., LinuxCNC, Mach3, or the machine's built-in controller).

(CAD) - Computer Aided Design

(formerly - Cardboard Aided Design :)

- is a way to digitally create 2D drawings and 3D models of real-world products before they're ever manufactured. With 3D CAD, you can share, review, simulate, and modify designs easily, opening doors to innovative and differentiated products that get to market fast.
- CAD drawing is a digital representation of an object, structure, or system created using specialized software to create precise and detailed visual plans. CAD drawings, utilized across various industries, such as architecture, engineering, and manufacturing, enhance accuracy and efficiency in the design process by streamlining development and production workflows.
- The most globally used proprietary CAD software is Autodesk-AutoCAD mainly for 2D drawing. However there are many other well known and widely used software such as (Dassault Systemes, Siemens, and PTC) in addition to multiple FOSS (Free and Open Source Software) tools such as, FeeCAD & OpenCAD.

(CAM) – Computer Aided Manufacture
(Computer Aided Machine)

- Involves using computers to control machines during the production of goods. By using CAM, designs can be sent to CAM machines such as laser cutters, 3D printers and milling machines. Design files must first be drawn up using computer aided design (CAD) software.

(CAE) - Computer-Aided Engineering

- is the use of computer software across a wide range of industries to simulate physics-based performance to improve product designs or assist in the resolution of engineering problems.

CNC Components

CNC vs Lathe

CNC: (Router vs Mill) Machining

- Milling machines are ideal for applications where precision is critical. CNC Routers: Generally produce lower cutting forces and are less precise in comparison.

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Thank You